



A synthesis of bird biodiversity studies in geographically isolated wetlands reveals research synergies and opportunities

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Abstract

Geographically isolated wetlands represent integral foraging and breeding habitat for many bird species across the United States, yet few studies have attempted to synthesize existing research to quantify generalizable relationships between bird biodiversity and geographically isolated wetlands. We conduct a systematic review of bird biodiversity in geographically isolated wetlands of the continental United States to explore the taxonomic and geographic areas of research and quantify the overarching themes in the literature. Most studies on bird biodiversity in GIWs come from the Midwestern United States, and only a third of all studies measured species richness and/or abundance for the entire avian community, with two thirds considering only a single taxonomic group (e.g., waterbirds) or a few key species (e.g., Black Tern, Northern Harrier, etc.). We find support from 15 US states for the species-area relationship for birds in geographically isolated wetlands. Semi-permanent wetlands (hydroperiod > 9 months) support higher species richness of birds compared to seasonal or permanent wetlands, albeit most studies came from the Midwestern United States. Future research of bird biodiversity in geographically isolated wetlands is needed outside of the Midwestern United States, especially in high wetland cover areas like Florida or countries like Brazil and Botswana. Studies should also examine biodiversity metrics (e.g., phylogenetic or functional diversity) beyond abundance and species richness which do not capture the multifaceted nature of biodiversity. Long-term temporal and spatial analyses of community composition and organization are also needed to determine if bird communities are homogenizing through time or supporting especially unique communities across space.

Keywords Synthesis · Review · Birds · Biodiversity · Wetlands · Geographically isolated wetlands

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Introduction

Wetlands are vital habitats for wildlife that occur at the interface between terrestrial and aquatic systems, accounting for over a third of rare and endangered species in the United States alone (US EPA 2015). Yet, wetlands have undergone considerable losses estimated between 20 and 35% in the last 300 years (Dixon et al. 2016; Fluet-Chouinard et al. 2023). Many of these lost wetlands come in the form of isolated or geographically isolated wetlands (hereafter referred to as GIWs), due to land conversion for housing and agriculture across the world (McCauley et al. 2013; Leibowitz 2003). GIWs are wetlands that are completely surrounded by upland habitat that lack a direct surface water connection for all, or some portion, of the year (adapted from Tiner 2003). GIWs have recently been revoked of federal protections under the Clean Water Act due to their lack of surface water connection, highlighting an urgent need for greater biodiversity and conservation attention as they continue to be lost or destroyed (“Revised Definition of ‘Waters of the United States’; Conforming” 2023). Documenting biodiversity and biodiversity loss in wetland ecosystems, tends to be concentrated in large biodiversity hotspots or coastal wetlands like the Everglades in Florida, the Okavango Delta in Botswana, or the Pantanal in Brazil, that contain high rates of endemism and species diversity (Junk et al. 2006; Ma et al. 2010). But, GIWs are also especially important for wetland biodiversity (Comer et al. 2005) and they are widely distributed, representing an estimated 16% of total wetland area in the United States alone (Lane and D’Amico 2016; Lane et al. 2023). Given their large spatial distribution in the United States, their removal of legal protections, and their continued loss, synthesizing the existing biodiversity patterns in these systems is imperative.

GIWs have unique hydrologic conditions because they lack surface water connection to other wetlands, creating ideal conditions for herpetological species, herbaceous plants, and higher dispersal organisms like birds. Birds are important biodiversity indicators (Oertli and Parris 2019), are well-documented taxonomically, and their use of and reliance on many wetland habitats makes them an excellent taxonomic group to understand biodiversity patterns in GIWs. Waterbirds rely on these ecosystems for various stages of their annual biology and survival (e.g., breeding, migration) (Forcey et al. 2011; Paracuellos 2006). Their importance to waterbird biodiversity has been studied most intensely in the Prairie Pothole Region of the midwestern United States, where many waterbird species use them as migratory staging and breeding habitat (Elliott et al. 2020; Fairbairn and Dinsmore 2001; Kantrud and Stewart 1977). GIWs may also act as satellite habitats for a variety of species to meet their life-history, especially for foraging as water levels fluctuate to match various prey and niche widths of a wide array of species (McKinney and Paton 2009). Although additional research on birds in GIWs has been conducted outside of this region (Barratt Heitmann et al. 2025a, 2025b; Czapka and Kilgo 2011; McKinney et al. 2011), it is unclear if similar biodiversity questions have been investigated in different parts of the United States. For example, determining support for the species-area relationship in different ecosystems, impacts of anthropogenic land cover at different spatial grains and extents, and the impacts of environmental filtering due to variable wetland hydrology. Furthermore, existing studies in GIWs have received considerable attention for wading birds and waterfowl, but it is unclear how, and to what extent, these patterns hold for other taxonomic groups (e.g., passerines) that are equally, if not more important for maintaining biodiversity.

Systematic reviews and syntheses can offer more generalizable conclusions through research weaving and collation of multiple, heterogeneous studies that have focused on different spatial and temporal units of analysis and geographic locations (Lortie 2014; Nakagawa et al. 2019; Spake et al. 2022). Systematic reviews for GIW biodiversity have focused extensively on certain anuran and herpetological species (Burrow and Lance 2022; Hamer and McDonnell 2008), but these systems are also heavily utilized by various avifauna across the continental United States. Another challenge in the review of GIWs studies is that current definitions and typologies associated with GIWs are primarily North America-centric. Nevertheless, the variety of terminologies that could characterize GIWs is numerous – e.g., over 35 different names as shown in Tiner (2003) alone. Such inconsistency and lack of clarity in GIW terminology can create research silos that overlook relevant research and lead to inaccurate or even misinterpretations of biodiversity patterns across the plethora of isolated wetlands in the United States and potentially the globe (Junk et al. 2006). Although global in distribution, nomenclature and wetland typology descriptions are only available for the United States. In order to review these systems globally, additional research is needed to identify the various names and wetland descriptions for these systems outside of North America—an objective outside the scope of this review. This nomenclature can silo different bird biodiversity studies in GIWs, highlighting the need for a more inclusive investigation of bird biodiversity studies in these systems more broadly.

Here in this study, we present a synthesis and systematic review of bird biodiversity studies conducted in GIW habitats from the continental United States. Our overall goal was to synthesize the bird biodiversity-GIW literature to date and provide an overview of gaps in this literature. To achieve this, we address three key objectives. Our first objective was to assess what bird species or groups of birds have been studied and where these studies have been conducted in the continental United States. Our second objective was to investigate which metrics of bird biodiversity have been measured (e.g., abundance, species richness, diversity, evenness) and determine whether these biodiversity measurements are consistent among species or taxonomic groups of birds and across the continental United States. Lastly, our third objective was to identify the emerging main thematic questions researchers have asked in this bird biodiversity-GIW literature.

Methods

We conducted our literature search and screening following the PRISMA protocol (Page et al. 2021). Specifically, we used 2 sets of keyword strings for literature searches on Web of Science and Scopus, on June 10, 2024 (Fig. 1). The first topic search conducted on each of the aforementioned databases, screened articles by keywords, article titles, and abstracts. Our first search had to include 1 word from each of 3 sets of nested terms: (“isolated” OR “ephemeral” OR “geographically isolated” OR “temporary”), (“pond” OR “wetland” OR “marsh” OR “depression”), and (“bird” OR “avifauna” OR “waterfowl” OR “passerine”). We also conducted a secondary topic search on both databases, that targeted explicit types of GIWs from Tiner 2003; and had to contain 1 word from each of 2 sets of nested terms: (“pot-hole” OR “rainwater basin” OR “sandhill” OR “salt flat” OR “salt lake” OR “desert spring” OR “kettle-hole” OR “Carolina Bay” OR “Cypress dome” OR “sinkhole” OR “vernal pool” OR “interdunal” OR “intradunal” OR “bog” OR “geysers” OR “rock pool” OR “seepage

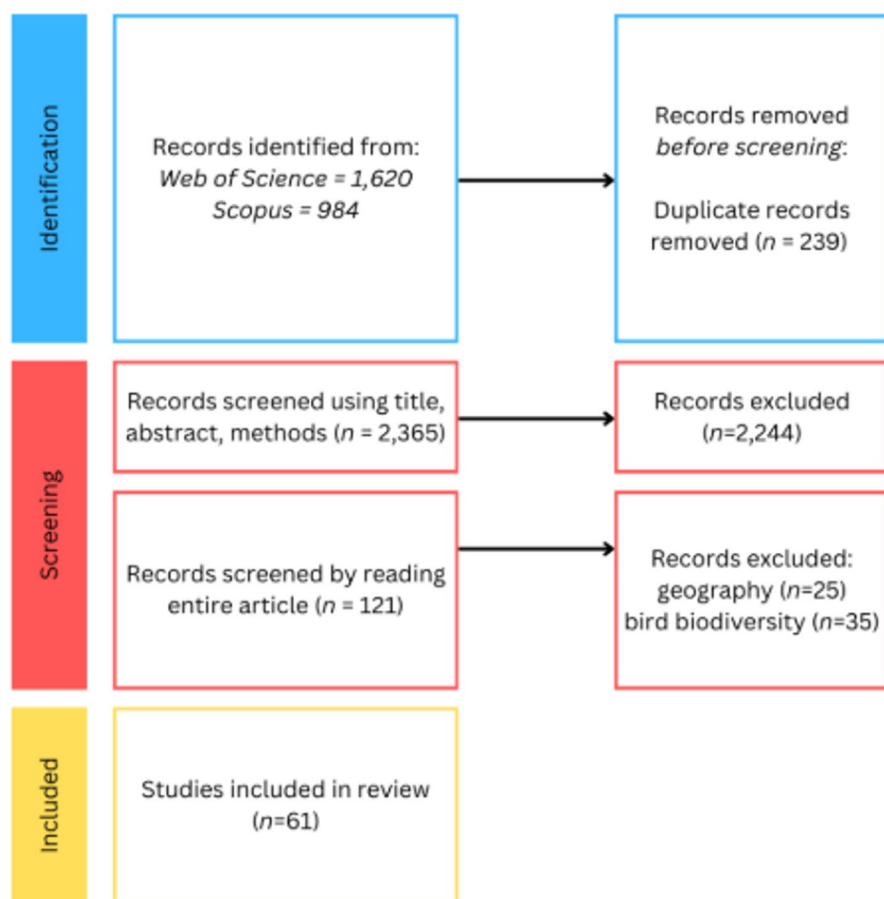


Fig. 1 Path diagram using PRISMA standards for publication screening and inclusion

slope” OR “fen” OR “depression” OR “excavated pond” OR “tarn” OR “excavated” OR “inactive floodplain” OR “Alvar”), and (“bird” OR “avifauna” OR “waterfowl” OR “passerine”). We performed multiple keyword searches to generate the greatest number of publication search results. Instead of restricting search terms to only GIW-specific wetland types found in Tiner (2003) (our second search), we included more general GIW related search terms to remove temporal bias for publications published after Tiner (2003) and Leibowitz (2003) using the aforementioned terminology (our first search).

We downloaded a total of $n = 1,620$ articles from Web of Science and $n = 984$ from Scopus, totaling $n = 2,604$. After merging these datasets and removing duplicates, we obtained a total of $n = 2,365$. We then began our screening process, where we removed all publications that did not include an explicit measurement of bird biodiversity (i.e., abundance, density, species richness, diversity) and/or was not conducted in at least one GIW by reading the title, abstract, and methods. This reduced the number of articles to $n = 121$. We then performed a second screening of these articles, where we read the entirety of the publication to ensure the aforementioned requirements were met, and also removed publications that

conducted research outside of the continental United States, since our definition of geographically isolated wetland by Tiner (2003) could not be easily applied to wetlands internationally or those in Alaska and Hawaii. GIWs represent a suite of different types of wetlands that may have different hydrologic, vegetative, and human-manufactured characteristics. In an attempt to be as inclusive as possible of all these different wetland types, we adapted our definition directly from Tiner (2003), where the wetland needed to be surrounded by upland habitat and lack direct surface water connection for all, or part, of the year. This includes seasonally pulsed wetlands adjacent to permanent water bodies like lakes and streams, man-made stormwater wetlands, or catchment wetlands that only have below-ground water transportation systems. These different wetlands are included because our birds are not able to travel via underground engineered hydrologic components, and so these still meet our definition of being surrounded by upland and lacking direct *surface* water connection.

We applied this definition to descriptions and/or images by the authors in the methods sections used to describe their study sites.

Of the $n = 121$ articles, in our first screening, we removed $n = 25$ for their presence outside the continental United States. We removed these because we did not feel confident that we had adequately sampled for the many types of GIWs globally with our list of United States-centric wetland search terms. We also removed an additional $n = 35$ publications that failed to either address bird biodiversity as a response variable, or the study was not explicitly conducted in at least one type of GIW. After this second screening, our database resulted in total of $n = 61$ studies of bird biodiversity in GIWs which were considered for our synthesis (Table S1). Hereafter we will refer to this as the *bird biodiversity-GIW literature*.

We then subsequently extracted data from each individual article. This included the study location (state), biodiversity measures collected/calculated, GIW type (if referenced by author), taxonomic unit of interest (waterbirds, passerines, etc.), and any themes in the analysis. Themes were developed during the review of each article after reading the methods and results, and were not compiled a priori. Themes were assigned based on the research objectives and questions from the authors in the study. These themes therefore were not mutually exclusive and were determined by screening the authors defined research questions in the introduction and subsequent results. Following the review of all articles, we grouped publications into general thematic categories based on the questions addressed by authors (Fig S1). After reviewing all publications, we had $n = 27$ identified themes, but in our results and discussion we only report on themes that had at least $n = 5$ articles (Table 1). Articles could fall into more than 1 theme, if for example the authors measured differences in bird species richness between wetlands and surrounding uplands, as well as those along a rural-urban gradient (e.g., McKinney and Paton 2009). Therefore, the total number of publications per thematic category are not mutually exclusive and do not add up to $n = 61$ in the results section.

Counting methodology

Given the sample size and heterogeneity in the datasets, we used a counting methodology (instead of calculating effect sizes using response ratios) to count occurrences for our 3 research objectives. For our first two objectives we simply tallied the number of times a biodiversity metric was collected and/or calculated and in which US state(s) the study was conducted in. For our final objective we counted the number of times an identified theme was

Table 1 Themes with more than 5 publications in the bird biodiversity-GIW literature and their respective descriptions. A full list of all publications in our final dataset ($n=61$) and their associated themes are provided in Table S1

Themes	Description	Example publication
Wetland Area	Bird counts and/or analyses were compared between GIWs with varying areas	Mamo and Bolen (1999)
Hydrology	Bird counts and/or analyses were compared in GIWs with varying hydrology, in either naturally occurring or experimentally manipulated systems	Kantrud and Stewart (1977)
Wetland Type	Bird counts and/or analyses were compared between different types of GIWs or GIWs were compared to other types of wetlands more broadly	Herteux et al. (2020)
Vegetation	Bird counts and/or analyses were compared between predominant forms of vegetation in GIWs or surrounding GIWs including tree species identities and dominant vegetation types	Keppeler et al. (2016)
Local and Landscape Land Cover	Bird counts and/or analyses were compared between local conditions surrounding and/or within GIWs and their landscape land cover context (continuous)	Anderson and Rooney (2019)
Restoration	Bird counts and/or analyses were compared between GIWs in natural (degraded) and restored states, or across some restoration gradient	Vanausdall and Dinsmore (2020)
Temporal	Bird counts and/or analyses were compared between seasons or across years (temporal) in GIWs	Tsai et al. (2012)
Climate	Bird counts and/or analyses were compared or modelled by abiotic climatic conditions, or projected climate change scenarios in GIWs	Steen and Powell (2012)

addressed by the authors in each study. Using the counting methodology allowed us to easily quantify the total number of occurrences of each theme and then discuss the overarching findings. This approach allowed us to gauge overall results and research implications more broadly, rather than using a topical modelling approach to extract certain words or phrases from manuscripts—which can narrow the interpretations of results or make publications difficult to compare more qualitatively. We therefore do not make any quantitative suggestions of significance, only suggest what signals have generally been found and emerged in the context of each theme. Compared to formal meta-analyses, one possible caveat for this method is that we do not extract effect sizes and the strength of each analytic technique is weighted equally regardless of the statistical modelling technique employed by the authors (Allen 2017). Within each theme, we counted how the bird biodiversity metric was affected by the associated theme. For example, if a publication found that agricultural cover was negatively associated with species richness at the landscape scale, that publication received a count of $n=1$ for “agriculture landscape negative”. This made comparisons between publications within the same theme simpler qualitatively albeit less stringent quantitatively, to better assess overall findings in the literature (see Fig S2-S9).

Results

Bibliometric summaries

After screening $n=2,604$ articles, we had $n=61$ (2%) that met our review criteria for being about bird biodiversity in isolated wetland(s) in the continental United States. These articles

came from $n=27$ different journals, *Wetlands* had the greatest number of articles with $n=12$, followed by *Journal of Wildlife Management* with $n=10$, but the median number of articles per journal was $n=1$ (Fig. 2). Publications ranged in publication date from 1953 to 2023, with a 10-fold increase in number of publications since 1970 (Fig. 2).

Studies came predominantly from the Prairie Pothole Region ($n=37$) located in the Great Plains which includes Iowa ($n=15$), Montana ($n=8$), Nebraska ($n=8$), North Dakota ($n=20$), South Dakota ($n=18$), and Minnesota ($n=12$) in the United States (Fig. 3). All of the lower 48 states had at least 1 publication, however, $n=13$ of these came from a single study (Dertien et al. 2020).

The most common reported measurements and metrics by authors across all studies was abundance ($n=61$), followed by species richness ($n=41$), density ($n=22$), some measurement of diversity (Shannon-Wiener Index, Jaccard's Index, Sorensen Index) ($n=7$), and presence/absence ($n=6$) (Fig. 2).

About half of studies counted all species within the community ($n=27$), while others were more taxonomically focused/restricted (Fig. 2). Of the more taxonomically constrained studies, most were interested primarily in waterbirds ($n=22$), although some studies focused even more specifically on this group and restricted their analyses to only waterfowl ($n=4$), shorebirds ($n=4$), or marsh birds ($n=4$) (Fig. 2). The remaining studies had two or more

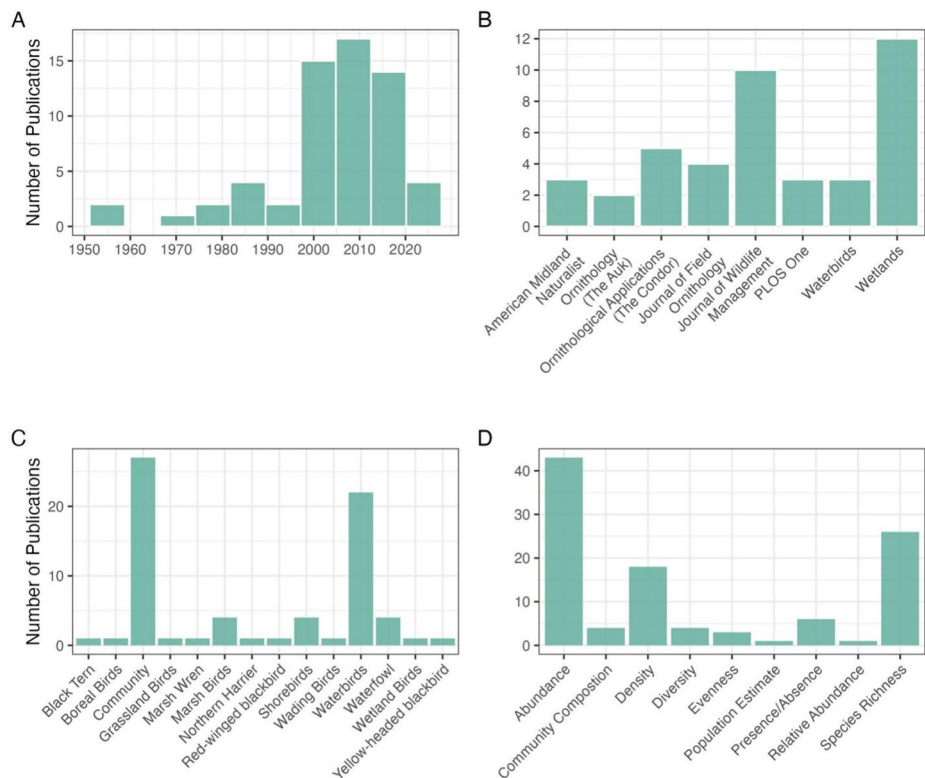


Fig. 2 Summary information for number of publications by **A)** year, **B)** journal, **C)** taxonomic unit of interest, and **D)** biodiversity metric collected

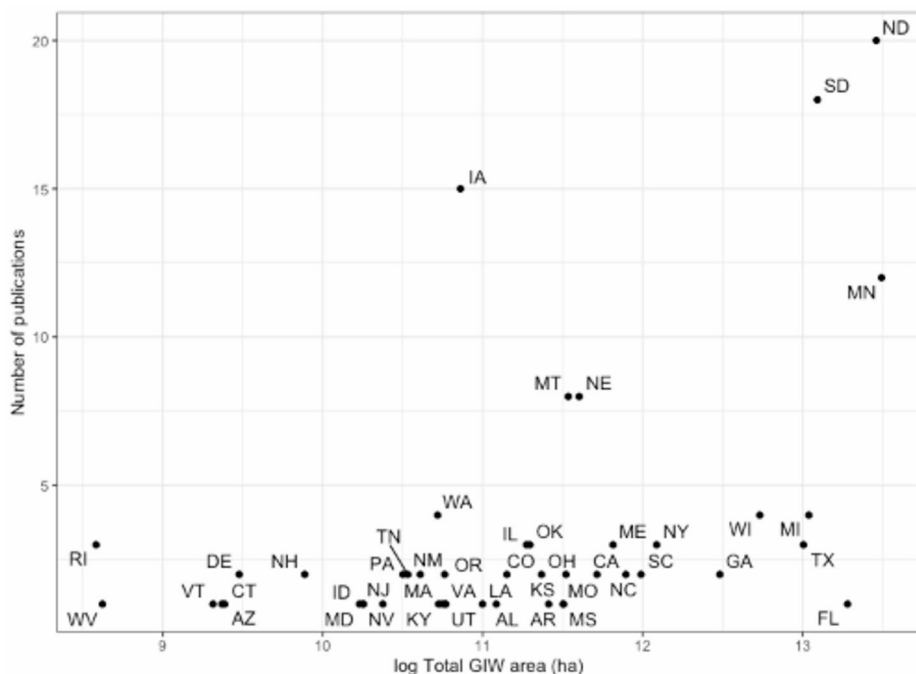


Fig. 3 Number of publications by US state and their respective GIW areas (Lane and D’Amico 2016). If studies were conducted across multiple states, each state was counted and weighted equally. States with 1 publication all come from a single study at the continental scale (Dertien et al. 2020)

specific focal species specifically, including Black Tern ($n=1$), Marsh Wren ($n=1$), Northern Harrier ($n=1$), Red-winged blackbird ($n=2$), and Yellow-headed blackbird ($n=2$).

Thematic representation

We identified a total of $n=26$ themes in the bird biodiversity-GIW literature (Fig S1). Of these, $n=8$ themes had more than five articles (Fig. 4). Of the $n=8$ themes that met our criteria of having more than 5 papers, the most common theme was local and landscape land cover ($n=23$), followed by wetland area ($n=12$), hydrology ($n=11$), climate ($n=7$), temporal ($n=6$), vegetation ($n=6$), wetland type ($n=6$), and restoration ($n=6$) (Fig. 4).

With the caveat that authors defined spatial scales quite variably, species richness and/or abundance were positively associated with higher wetland cover at the landscape scale in $n=9$ (9/23, 39%) of these studies. An additional $n=4$ (4/23, 17%) publications however found that increased wetland cover at the landscape scale was associated with lower species richness or density of at least 1 species in the study. Surprisingly, $n=4$ (4/23, 17%) publications found a positive relationship between urbanization cover at the landscape scale and species richness (Eakin et al. 2018; McKinney and Paton 2009) or abundance of a single species (Ballard et al. 2021; Saunders et al. 2019). Only a single study (1/23, 4%) found a negative relationship between urbanization cover at the landscape scale and abundance of half the study species (Saunders et al. 2019).

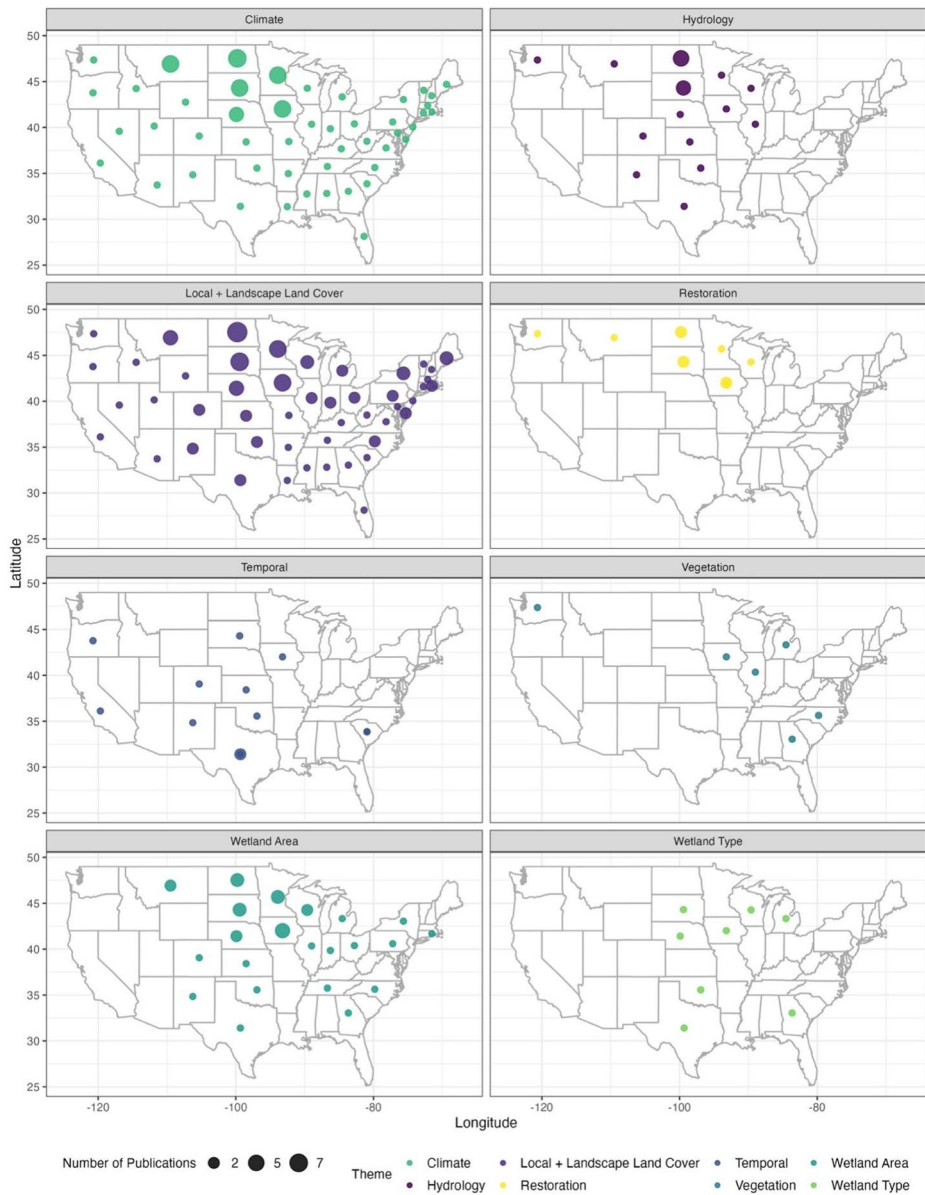


Fig. 4 Number of publications per theme from each state in the continental United States. Only publications containing themes that had > 5 publications are displayed ($n=8$)

A majority of studies (16/19, 84%) found that increasing wetland area was associated with increased abundance ($n=9$), density ($n=1$), species richness ($n=10$), occupancy ($n=1$), and/or diversity ($n=1$). $N=7$ publications quantified impacts of climate in the bird biodiversity-GIW literature, including projected changes due to climate change and annual effects of precipitation on abundance and species richness. Three publications (3/7, 43%)

found significant species range contractions due to climate change projections including highly variable annual precipitation and drought severity (Sorenson et al. 1998; Steen and Powell 2012; Steen et al. 2014). The other four publications (4/7, 57%) in the climate theme modelled abundance and species richness changes due to abiotic climatic conditions, including precipitation (Forcey et al. 2007, 2011; Niemuth and Solberg 2003) and temperature (Dertien et al. 2020).

Of the $n=11$ papers in the hydrology theme, most (8/11, 72%) investigated different defined hydrology in the PPR region of the United States, where semi-permanent wetlands often had the highest species richness, compared to seasonal and permanent wetland classes. In the temporal theme, all of the studies looked at seasonal changes in bird species richness and abundance, with half (3/6, 50%) considering the changes across the full annual cycle, and the other half only considering the Spring, Summer, and small portions of the Fall. In the wetland type theme, three of the publications (3/6, 50%) compared bird biodiversity estimates between agricultural (man-made) wetlands to natural wetlands, and another three compared bird biodiversity between GIWs and other wetland types.

In the vegetation theme, half of the papers (2/6, 33%) found that emergent vegetation increased abundance for waterbirds, and half (3/6, 50%) found that trees, or species-specific tree species, increased species richness. Of the four publications in the restoration theme analyzing differences in richness, abundance, and density of birds in natural compared to restored wetlands, (2/4, 50%) found that restored wetlands had higher estimates (Creighton et al. 1997; Elliott et al. 2020), (1/4, 25%) found that natural wetlands were higher (Delphey and Dinsmore 1993), and (1/4, 25%) found no difference (Ratti et al. 2001).

Discussion of top identified themes

The themes and aims of the publications we identified in this review are not mutually exclusive, as vegetation can also be a component of local and landscape land cover comparisons, as is hydrology and wetland type. Therefore, although we break these different themes up into separate sub-categories, we do recognize that there are cross-cutting themes in some of the results. Due to our counting method, we recognize that equal weight is being attributed to each publication regardless of how strong the evidence was in the results of the publication (Game et al. 2018). However, we only count a theme's reported pattern if the test was deemed statistically significant by the author. Themes and the number of publications that were included in that theme are reviewed in turn below.

Local and landscape land cover ($N=23$)

The scale and buffers at which authors defined local and landscape land cover varied across publications; local land cover scale varied from the attributes of a single wetland to 1 km radius around wetlands (Eakin, Hunter, and Calhoun 2018; McKinney and Paton 2009) and landscape land cover scale varied from 3 km to EPA Level II ecoregion (e.g., Mojave and Sonoran Desert Plains) (Dertien et al. 2020; Fairbairn and Dinsmore 2001; Forcey et al. 2007, 2011). For example, Webb et al. (2010) used a 25 km buffer to infer an overall negative species richness relationship to landscape-scale wetland cover in Nebraska, while Fairbairn and Dinsmore (2001) used a 3 km buffer and found a strong positive relationship

between total species richness and wetland landscape land cover in Iowa. Both of these studies examined wetland-dependent waterbird and shorebird species, which may indicate that foraging distance and/or dispersal range is impacting habitat use variably between the scales of 3 km to 25 km. However, it is important to note that buffer distance choices are not consistent amongst studies, likely reflecting arbitrary choices by the authors, rather than reflecting a true ecological mechanism. Future studies need to test multiple buffers to ensure they can find the true scale of effect (Barratt Heitmann et al. 2025a, 2025b).

Of the four studies that quantified the impacts of urbanization cover most found a positive relationship between urbanization and abundance or species richness. Three found a positive relationship with at least 1 species, while only 1 study found a negative relationship. This points to the value of wetland habitats in urbanized landscapes which can provide important biodiversity value, and potentially higher value than in rural settings due to increased urban tolerance in select bird species and/or reduced wetland cover creating fewer habitats throughout the landscape (Callaghan et al. 2021). However, of the three publications which found this positive association, 2 were only documented in 1 species' abundance (American Coot and Marsh Wren) in Minnesota and North Dakota with low urbanization footprints, and a third only found this positive relationship in Maine in urban-tolerant species richness. These important caveats mean that urbanized land cover may not act uniformly across all landscapes and ecotones, and that environmental filtering of the communities utilizing wetlands are not one-size-fits-all.

Lastly, total species richness was positively associated with increased local vegetation cover within GIWs in Webb et al. (2010) and Fairbarin and Dinsmore (2001), while Naugle et al. (2000) found a significantly positive relationship for only some species of waterfowl. Ballard et al. (2021) found that dabbling duck abundance was positively associated with emergent vegetation cover and diving duck abundance with submerged vegetation. All four studies that found a significant positive relationship were from the PPR region, and $\frac{3}{4}$ only considered waterbirds/waterfowl—not the entire community. Nonetheless, it appears that emergent vegetation at the local scale can play an important positive role in increasing both abundance and richness in GIW habitats for birds. This likely indicates that restored and created wetlands should leverage both flooded vegetation or submerged aquatic vegetation to promote bird habitat use, especially in manicured and mowed urban stormwater ponds that lack suitable vegetation characteristics to support biodiverse bird communities.

Wetland area (N = 19)

This indicates that these habitats are increasingly more valuable the larger they are, contrary to other wetland habitats like lakes which don't always show similar patterns of increased abundance or density with area (Ballard et al. 2021; Ruwaldt et al. 1979). Isolated wetlands therefore conform generally to the species-area relationship indicating that larger wetlands tend to support more individuals and species than smaller ones, possibly due to increased wetland edge for foraging, a greater variety of niches, or generally greater prey availability (Barratt Heitmann et al. 2025a, 2025b; Cerda-Peña and Rau 2023). Only two found no relationship, and only a single study found that wading bird density was negatively associated with increased wetland surface area. These studies may have not found a positive relationship due to isolated wetlands being too similar in size, such that the gradient was not large enough to detect a significant difference (e.g., McKinney and Paton 2009; Scheffers et al.

2006). The only publication to find a negative result was attributed to lower water depths in smaller wetlands that made foraging much easier for wading birds compared to larger excavated wetlands with large water depths and therefore inadequate foraging habitat (Herteux, Gawlik, and Smith 2020). The majority of papers suggest that wetland area plays a key role in increasing both abundance and species richness in birds, and when choosing to conserve or preserve wetlands, larger GIWs will typically have higher biodiversity value to birds.

Hydrology (N = 11)

The hydrologic regimes in wetlands, including wetland depth and hydroperiod, have long been studied for their impacts to biodiversity and birds are sensitive to changes in hydrologic condition of wetlands for foraging, nesting, and evasion of predators. Many of the papers that compared the impacts of different wetland hydroperiod (water permanence) occurred in the Prairie Pothole Region, where a robust system of wetland ranking based on hydrology exists from ephemeral (IV) to permanent (I). These publications consistently found that semi-permanent wetlands (II) had the highest estimates of species richness and abundance compared to all other wetland hydrologies (Austin 2002; Kantrud and Stewart 1977, 1984; Ruwaldt et al. 1979; O'Neal et al. 2008). Semi-permanent wetlands often do not dry up every year, meaning that they seem to retain ideal hydrologic conditions for a large variety of avifauna to forage, promoting sustained emergent and aquatic plant growth due to lower hydrologic stress from annual drying. Additionally, these semi-permanent wetlands may offer more spatial and temporal heterogeneity of water permanence and vegetation, that meet expands the niche space for a greater number of species. have higher Seasonal (III) and ephemeral (IV) wetlands however also showed proportionally high species richness and abundance even with relatively less wetland area, but were subject to more year-to-year variation due to being shallow, precipitation-fed wetlands (Kantrud and Stewart 1984; Ruwaldt et al. 1979). Many of the aforementioned publications were primarily concerned with waterbird species use of these wetlands, other studies from outside the PPR region found that marsh birds had varying response to water levels in GIWs; e.g., American Bitterns responded positively to increased water depth, and marsh birds in the Great Lakes coastal wetlands responded positively to higher water permanence (Gnass Giese et al. 2018; Jorgensen et al. 2022). Furthermore, wetland hydroperiod manipulation in Illinois had a positive effect on abundance and richness for shorebirds and waterbirds compared to wetlands with little or no water management (O'Neal et al. 2008). More permanent wetland hydrology in permanent and semi-permanent wetlands is generally associated with higher species richness and abundance of most bird species in GIWs, but those wetlands with more ephemeral hydrology can still contribute greatly to species richness in high precipitation years, often having higher species turnover than their more permanent counterparts (Daniel et al. 2019). Land managers looking to increase species richness and abundance should consider managing GIWs for increasing water permanence, especially in drought years when wetland availability on the landscape can be quite low. Another potential implication to consider from this review, is that the wetland hydrologic regimes are well-documented in the PPR region, but hydrologic regimes in other parts of the United States may vary considerably. For example, vernal pools (a type of GIW) in the Northeastern United States tend to be very ephemeral with no semi-permanent hydrologies (Tiner 2003). Therefore, the spatial

configuration and arrangement of these hydrologies spatially may not mimic that of the PPR region, where they occur in high densities, and close spatial proximity for birds to exploit.

Climate (N=7)

Climatic conditions and abiotic variables influence GIWs specifically, due to being primarily precipitation fed across much of their continental distribution (Tiner 2003). In some cases, modelled species ranges and abundances were eliminated entirely in increased atmospheric CO₂ climate conditions (100%: Black Tern) while others experienced only moderate range reductions (29%: American Bittern) (Steen et al. 2014). Higher soil moisture, previous year precipitation, and same year precipitation were all positively associated with higher duck abundance in the Prairie Pothole Region and the Northern Great Plains (Forcey et al. 2007, 2011; Niemuth and Solberg 2003) and at the continental scale higher temperature (i.e., lower latitudes) were associated with higher species richness (Dertien et al. 2020). Importantly, Dertien et al. (2020) did not simulate potential temperature changes due to climate change within respective latitudes, suggesting rather, that this is more likely a product of higher biodiversity closer to the equator (Brown 2014). Publications which investigated the impacts of precipitation found higher abundance associated with higher same year and previous year precipitation. This means that the bird biodiversity in GIW systems are highly correlated with increased rainfall, that likely lead to higher reproductive success due to higher food availability and more suitable nesting areas, especially in the Prairie Pothole Region. However, higher rates of endemism and richness in southern regions of the United States (Dertien et al. 2020) may indicate that greater attention is needed to understand the implications of climate on birds in GIWs, including temperature fluctuations at a single site, especially because waterbirds and waterfowl were the taxonomic units considered by all but one publication in this theme.

Temporal (N=6)

Changes in community composition or biodiversity seasonally, or across years has become a central tenet of many biodiversity studies aiming to document population declines or variation in species occupancy through the year. However, in our review only six publications investigated this theme, with most considering only seasonal changes and only a single study comparing long-term changes over multiple years. California vernal pools had the highest abundance of waterbirds present in early spring prior to migration northward, similar to Carolina Bays in South Carolina that had higher species richness and abundance in the winter months prior to migration (Bogiatto and John 2006; Czapka and Kilgo 2011). Long-term historical studies of bird habitat use in Oregon from the 1800s to present suggests there has been a decrease in 4 particular species of waterbirds (Taft and Haig 2003). The number of breeding duck pairs was higher in playa lakes in Texas in May compared to June, highlighting their importance early in the breeding season (Ray et al. 2003). Hartzell et al. (2007) found that the proportion of wetland-dependent species was highest in the spring compared to all other seasons but the proportion of waterfowl species was similar in both spring and fall (Hartzell et al. 2007). A final study comparing ground count and aerial survey methods found that May aerial surveys in South Dakota occurred too early and overestimated the breeding duck population, and subsequently underestimated brood success in

most species due to late nesting attempts in late July and August (Naugle et al. 2000). In total, all studies found that GIWs had the highest abundance or richness of avifauna in the spring or winter. This suggests that although much attention has been paid to these systems as breeding habitat in summer months (Naugle et al. 2000), they may be of equal or greater importance as winter or spring migration staging habitat for waterbirds. Furthermore, this is evidence that wetland habitats with mild winter climates (South Carolina, California, and Texas) require attention outside of the breeding season, when they support the most species and highest abundance of migrating waterbirds. Lastly, although seasonal changes have received attention in this literature, only one single study documenting long-term temporal changes in biodiversity was identified in our review that wasn't directly measuring the impacts of restoration (see *Restoration* below). Future work on long-term temporal changes in these communities is warranted especially in the context of huge avifauna declines in North America (Rosenberg et al. 2019). One important consideration not directly addressed in these papers is also that precipitation that fills GIWs may be timed different latitudinally, and may also be shifting as climate change alters precipitation patterns seasonally, potentially changing when birds can utilize these wetlands. The implications of seasonal changes on birds in GIWs also appears to be highly context dependent, but current studies often lack true annual survey methodologies leaving gaps in the biodiversity utility of these systems outside of the typical breeding and migratory seasons.

Vegetation (N = 6)

GIWs are unique in that their vegetation and vegetative structural components beyond broad land cover categories (trees, shrubs, emergent vegetation), can vary widely depending on the type of wetland, land management, and hydrology (Tiner 2003; Herteux et al. 2020). Herteux et al. (2020) found that open agricultural ponds with little vegetation were used primarily during the early spring by wading birds, before shifting to Cypress-Gum ponds dominated by Bald Cypress and Black Gum in late spring and early summer. Similarly, Mamo and Bolen (1999) found that Carolina Bays embedded in forests with higher canopy cover had higher abundance and species richness of songbirds than those embedded in clear-cut pine plantations. In Michigan, Black Spruce Tamarack bogs harbored the highest abundance of songbirds compared to their open canopy counterparts (Ewert 1982). Two studies from the PPR region found that emergent vegetation in wetlands increased abundance of dabbling ducks and submerged aquatic vegetation increased abundance of diving ducks (Ballard et al. 2021; O'Neal et al. 2008). Studies investigating this theme came from 6 different states across the US, and all identified that wetland vegetation in the form of submerged vegetation, emergent vegetation, and shrubs or trees (at some spatial scale) increase abundance and richness of avifauna. We should note however, that we did not directly test how these vegetation attributes differed across geographies, so we can only comment on the general signal we found, not the relative importance of each within their given geographic contexts. This does suggest that open water may not be optimal for high biodiversity value, but rather that plant and tree species indicative of their respective geographies help promote biodiversity in GIWs. This is consistent with findings from the land cover section above, which indicates that environmental filtering of the bird community that use GIWs is not constrained at the landscape scale, but rather by the local vegetation attributes present at wetland sites. Critically, it means that vegetation characteristics at the wetland scale can be harmonized to

meet the foraging requirements of the greatest number of species, i.e., aquatic vegetation for diving ducks, emergent vegetation for dabbling ducks, and wetland dependent tree species for songbirds.

Wetland type (N = 6)

Of the publications in this review six compared bird biodiversity metrics at one or more types of wetlands, including among different types of GIWs and between GIWs and other wetland types (e.g., lakes or rivers). Playa wetlands, circular shallow depressional wetlands in deserts and prairie habitats, had higher density of duck pairs in Texas than their created counterparts (Ray et al. 2003), but agricultural wetlands had higher densities of wading birds than natural ones for half the year, and less than natural Cypress-Gum ponds in the other half (Herteux et al. 2020). In Oklahoma, depression wetlands and man-made wetlands did not differ significantly in species richness, but natural wetlands had higher evenness than created wetlands which showed high dominance by a single species (Hartzell et al. 2007). However, community composition was quite similar (*Jaccard's Similarity Index* = 67%) between natural and man-made wetlands (Hartzell et al. 2007). Of the remaining publications looking at wetland type differences, Ewert (1982) focused on bog type in Michigan, where open bogs had the lowest estimates of abundance and only contained relatively common species compared to White-Cedar and Black Spruce-Tamarack bogs (Ewert 1982). Gnass Giese et al. (2018) documented that inland GIW wetland types in Wisconsin acted as refugia for bird species in dry years, but that coastal wetlands around the Great Lakes had higher species richness in years with high precipitation. Finally, amongst all wetland types in Iowa, semi-permanent GIWs had the highest abundance (corrected for wetland area) compared to lakes, man-made wetlands, and seasonal GIWs (Ballard et al. 2021). Overall studies on wetland types yielded inconsistent results. Species richness and abundance don't seem to respond uniformly to created agricultural wetlands versus natural wetlands a variety of environmental contexts. This may indicate that there is either a temporal component driving habitat-use of man-made wetlands following vegetation succession, or the management regimes in created wetlands significantly impact the observed biodiversity compared to their natural counterparts.

Restoration (N = 6)

The impacts of wetland restoration on biodiversity compared to their natural counterparts has been of critical interest to restoration ecologists. In two studies that examined how restoration impacted the bird community over time, one study found that wetland-dependent bird density did not increase through time (Fletcher and Koford 2003). The other study found that over 10 years, abundance, richness, and density of birds in the entire community increased, likely due to vegetation succession, and high vegetation heterogeneity within individual restored wetlands (Hapner et al. 2011). However, Hapner et al. (2011) did find that wetland-dependent species richness decreased over the restoration period. Fletcher and Koford (2003) argued that restoration was not effective at increasing density, abundance, or species richness because of low precipitation in restored wetland habitats on former agricultural fields. Conflicting evidence for increased species richness and abundance of birds in restored wetlands and in time since restoration, indicates that management throughout

restoration including through plant installation and hydrologic control likely influences biodiversity responses to restoration. Studies that found either no response or a negative response in restored sites over time or compared to natural wetlands, had less stable hydrology compared to the studies which found a positive response to restoration. Of the studies in this theme, a few compared GIWs to their baseline year (2/6, 33%), while others compared restored wetlands to natural counterparts that were similar in size and vegetation (4/6, 66%). Importantly, the conflicting results of restoration on biodiversity from the studies may be due in part to these contrasting study designs and references, and the fact that restorations ranged in age from 1 to 16 years, but there was not an even number of age classes in all of these studies, since many of the restorations were carried out by Ducks Unlimited, and were outside the purview of the researchers. This suggests that restoration studies need to consider how the temporal scope of their study design impacts the species richness and abundance accumulation curves in both linear and non-linear ways.

Limitations of current review

An important limitation of this synthesis is that apparent differences among studies should not be interpreted as evidence of regional effects. Most studies differed simultaneously in wetland type, hydrologic regime, surrounding landscape, focal taxa, sampling methodology, and geographic location. Consequently, our synthesis identifies recurring ecological patterns across studies but cannot attribute contrasting results to geography itself. Future quantitative meta-analyses with sufficient replication across wetland types and regions will be needed to separate geographic effects from other sources of ecological variation. Similarly, extracting data from these papers to analyze the quantitative variation in biodiversity indices can shed light on which factors (land cover, hydrology, wetland vegetation, etc.) are most important, and what their quantitative relationship is to biodiversity (e.g., linear v non-linear). Lastly, our review here is quite restrictive in terms of terminology that needed to be present, and restrictive geographically due to our search terminology. GIWs are present globally and future work needs to consider how to incorporate the most inclusive set of search terms, especially in different languages, across the globe.

Synthesis and conclusion

Our review of the bird biodiversity-GIW literature has elicited a few key significant findings on the taxonomic units of study, the geographic scope of research, and overarching themes and biodiversity metrics that would inform future research avenues in this field. First, taxonomic representation is quite varied in the bird biodiversity-GIW literature. Only one third of the reviewed publications measured biodiversity for the entire bird community, while the remaining two thirds were more taxonomically restricted considering only a single taxonomic group of birds like waterbirds and waterfowl. Unsurprisingly, the geographic scope is also skewed to the Prairie Pothole region of the United States where ~2/3 of all studies were conducted because of their dense waterfowl breeding bird populations. Although not a central objective of this review, the term GIW in our searching has largely failed to coalesce the GIW literature more generally. This is evidenced by only a single publication (Hertuex et al. 2020) using the term GIW to describe these wetlands even though consider-

able attention has been paid to uniform naming conventions of these wetlands (Mushet et al. 2015; Leibowitz 2015). Lastly, we found that close to 2/3 of all the publications in this review measured only abundance and/or density, with the other 1/3 collecting species richness metrics. Fewer than 10% of all the studies in this review measured diversity as a metric in their studies, even though biodiversity is inherently multifaceted including functional and phylogenetic components not captured in any of the publications in this review and not captured holistically by abundance or species richness alone (Hillebrand et al. 2018; Htay et al. 2023).

We also identified some key takeaways from our thematic analysis of the bird biodiversity-GIW literature. We found strong support for the species-area relationship in GIWs, indicating that larger wetlands support a greater number of species. However, we should not discount the importance of smaller wetlands that play an outsized role in high precipitation years when seasonal wetlands show considerable increases in abundance and species richness of birds wetlands (Kantrud and Stewart 1984; Ruwaldt, Flake, and Gates 1979). Semi-permanent hydrologic conditions are the most biodiverse GIWs and support the greatest number of individuals (Austin 2002; Kantrud and Stewart 1984; 1977; Ruwaldt, Flake, and Gates 1979; O'Neal, Heske, and Stafford 2008). Importantly, this means that vegetative restoration in restored or created wetlands is not enough to increase abundance and species richness of birds, but instead requires a combination of hydrologic and vegetative restoration in concert (Hapner et al. 2011; Fletcher and Koford 2003). Temporal studies in this review showed that the non-breeding season may support the highest abundance and richness of species in GIWs, especially in ecoregions outside of the Prairie Pothole Region (Bogiatto and Karnegis 2006; Czapka and Kilgo 2011). Collectively, these studies suggest that bird biodiversity in GIWs is governed by a relatively small number of interacting ecological processes (e.g., hydrology, wetland size, vegetation, and landscape context) despite the wide diversity of wetland types and bird taxa studied.

In addition to our main identified themes, we also found a considerable number of additional themes that contained only a few publications (Fig S1). Of these, the chief themes with more than a single publication included those interested in documenting differences particularly in species richness of GIWs and their surrounding upland habitats. Of which, many documented higher richness in wetlands compared to their surrounding uplands (McKinney and Paton 2009; Scheffers et al. 2006; Czapka and Kilgo 2011). Additionally, authors were also concerned with changes between rural and urban GIWs (McKinney and Paton 2009), measurements of waterbird populations and their breeding success (Skagen et al. 2008; Sorenson et al. 1998), and older papers containing general natural history descriptions of bird and their abundances (Harris 1954; McKown et al. 2021).

The future of bird biodiversity-GIW research could be organized towards three main axes of knowledge gaps identified in this review: space, time, and taxa. None of the studies we identified in this review, to date, have considered the biodiversity value of GIWs continentally, or more importantly, at the global scale, where they may be more important at lower latitudes with higher precipitation. In light of our current biodiversity crisis and substantial ongoing wetland loss (Dixon et al. 2016; Fluet-Chouinard et al. 2023), it is critical to know if these wetlands are extraordinarily important for birds outside of the United States or if they're more or less valuable to non-waterbird taxonomic groups, which reflect only a fraction of all species with different functional and phylogenetic biodiversity values. Finally, none of the studies in this review have investigated temporal biodiversity trends in

depth, which can be challenging due to data constraints. As local, national, and international policies continue to change, it is important that future research works to identify long-term trends in richness, species turnover, or abundance that may indicate particular vulnerability to climate change and urbanization, ultimately highlighting the continued importance of GIWs for bird biodiversity.

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Author contributions J.B.H, C.T.C, and J.Q conceived and designed the study. J.B.H collected and analyzed the data. J.B.H wrote the manuscript, and all authors commented, revised, and approved the final manuscript.

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Declarations

Competing interests The authors declare no relevant financial or non-financial interests to disclose.

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